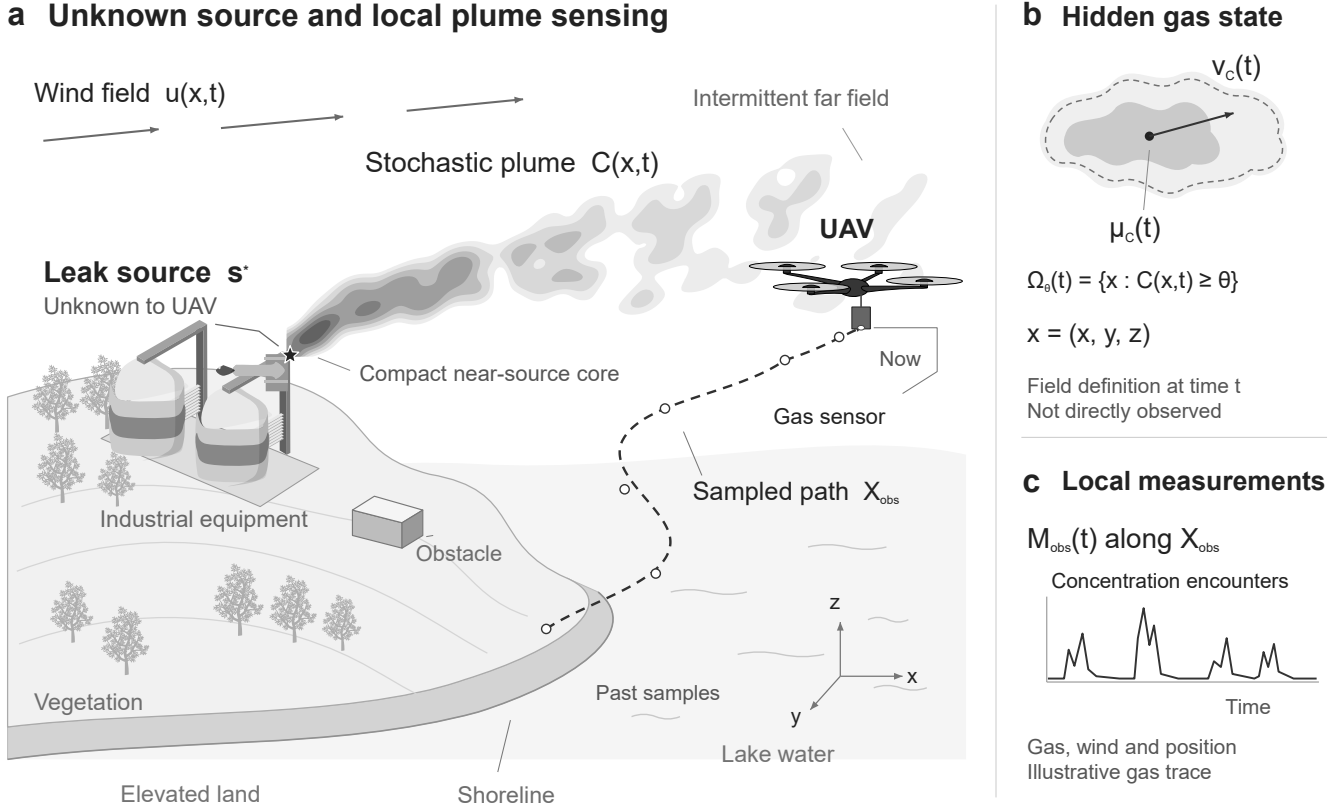

a Unknown source and local plume sensing
Wind field u(x,t)
Intermittent far field
Stochastic plume C(x,t)
UAV
Leak source s*
Unknown to UAV
Compact near-source core
Now
Gas sensor
Sampled path X_obs
Industrial equipment
Obstacle
Vegetation
Past samples
z
x
y
Lake water
Elevated land
Shoreline
b Hidden gas state
v_c(t)
μ_c(t)
Ω_θ(t) = {x : C(x,t) ≥ θ}
x = (x, y, z)
Field definition at time t
Not directly observed
c Local measurements
M_obs(t) along X_obs
Concentration encounters
Time
Gas, wind and position
Illustrative gas trace